



Entrega 2 Semana 5: Implementación de Jenkins en SkillCert

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Énfasis Profesional I - Integración Continua

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Resumen

Esta entrega implementa Jenkins como gestor de operaciones de integracion continua para SkillCert. El pipeline valida el repositorio con checkout desde GitHub, instalacion de dependencias, pruebas backend, build frontend, validacion de Docker Compose y construccion de contenedores.

Objetivo

Aplicar integración continua mediante Jenkins y demostrar con evidencias visuales que el pipeline se ejecutó correctamente.

Repositorio

<https://github.com/Jonevan369/skillcert-integracion-continua>

Características requeridas para Jenkins

Caracteristica	Implementacion
Jenkins	Jenkins LTS 2.541.3 ejecutado en contenedor temporal.
Node.js	NodeJS-20 configurado en Jenkins; consola evidencia v20.20.2.
Docker	Docker Desktop activo y socket Docker montado en el agente Jenkins.
Build tools	build-essential instalado para compilar dependencias nativas como better-sqlite3.
Plugins	Pipeline, Git, GitHub Branch Source, Pipeline Stage View, NodeJS, Docker Workflow y Timestamp.
Repositorio	GitHub: https://github.com/Jonevan369/skillcert-integracion-continua

Pipeline ejecutado

Checkout, Install Dependencies, Backend Tests, Frontend Build, Validate Docker Compose, Build Containers y Post Actions terminaron en SUCCESS.

Evidencias visuales

Captura 1. Dashboard Jenkins con job skillcert-ci y ultimo exito

**Jenkins**

+ Nueva Tarea

Historial de trabajos

Trabajos en la cola

No hay trabajos en la cola

Estado del ejecutor de construcciones

0/2

Todo

S	W	Nombre ↓	Último Éxito	Último Fallo	Última Duración
✓	☀	skillcert-ci	1 Min 57 Seg #1	N/D	1 Min 0 Seg

Icono: S M L

añadir descripción

añadir descripción

REST API

Jenkins 2.541.3

Captura 2. Build Jenkins exitoso con repositorio y duracion

**Jenkins**

skillcert-ci

#1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Git Build Data

Restart from Stage

Replay

Pipeline Steps

Workspaces

✓ #1 (6 jun 2026 23:13:50)

añadir descripción

Conservar esta ejecución para siempre

 **git**

Revision: 7c471d2484a411e04861eef2894ae8f71e8c6207

Repository: <https://github.com/Jonevan369/skillcert-integracion-continua>

• refs/remotes/origin/main

</>

Sin cambios

Comenzó hace 1 Min 58 Seg

Tardó 1 Min 0 Seg

REST API

Jenkins 2.541.3

Captura 3. Etapas del pipeline en SUCCESS desde Jenkins wfapi

Jenkins Pipeline - skillcert-ci #1 - SUCCESS

Duracion total: 60.0 s | Fuente: http://localhost:8080/job/skillcert-ci/1/wfapi/describe

Etapas	Estado	Duracion
Declarative: Tool Install	SUCCESS	0.07 s
Checkout	SUCCESS	4.10 s
Install Dependencies	SUCCESS	48.38 s
Backend Tests	SUCCESS	0.34 s
Frontend Build	SUCCESS	1.92 s
Validate Docker Compose	SUCCESS	0.32 s
Build Containers	SUCCESS	3.15 s
Declarative: Post Actions	SUCCESS	0.29 s

Captura 4. Consola Jenkins: npm test -w server y pass 3

Consola Jenkins - npm test -w server y 3 pruebas aprobadas

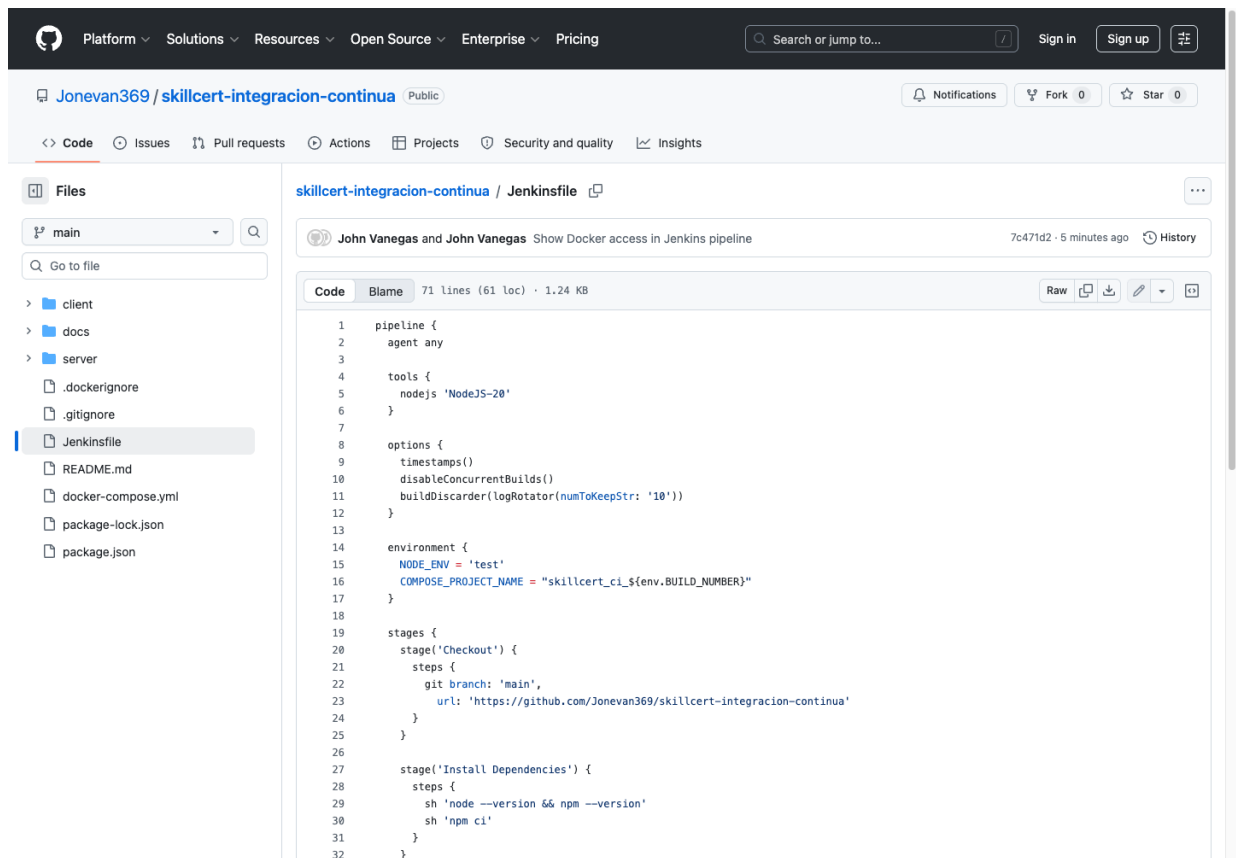
```
[Pipeline] stage
[Pipeline] { (Backend Tests)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
[2026-06-06T23:14:44.324Z] + npm test -w server
[2026-06-06T23:14:44.324Z]
[2026-06-06T23:14:44.324Z] > @competence-platform/server@0.1.0 test
[2026-06-06T23:14:44.325Z] > node --test src/tests/*.test.js
[2026-06-06T23:14:44.325Z]
[2026-06-06T23:14:44.325Z] TAP version 13
[2026-06-06T23:14:44.325Z] # Subtest: eligibleBadgesForSkill assigns badges at configured thresholds
[2026-06-06T23:14:44.325Z] ok 1 - eligibleBadgesForSkill assigns badges at configured thresholds
[2026-06-06T23:14:44.325Z] ----
[2026-06-06T23:14:44.325Z]   duration_ms: 0.878375
[2026-06-06T23:14:44.325Z]   ...
[2026-06-06T23:14:44.325Z] # Subtest: voteWeightForKarma caps reputation influence
[2026-06-06T23:14:44.325Z] ok 2 - voteWeightForKarma caps reputation influence
[2026-06-06T23:14:44.325Z] ----
[2026-06-06T23:14:44.325Z]   duration_ms: 1.658542
[2026-06-06T23:14:44.325Z]   ...
[2026-06-06T23:14:44.325Z] # Subtest: calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly
[2026-06-06T23:14:44.325Z] ok 3 - calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly
[2026-06-06T23:14:44.325Z] ----
[2026-06-06T23:14:44.325Z]   duration_ms: 0.11425
[2026-06-06T23:14:44.325Z]   ...
[2026-06-06T23:14:44.325Z] 1..3
[2026-06-06T23:14:44.325Z] # tests 3
[2026-06-06T23:14:44.325Z] # suites 0
[2026-06-06T23:14:44.325Z] # pass 3
[2026-06-06T23:14:44.325Z] # fail 0
[2026-06-06T23:14:44.325Z] # cancelled 0
[2026-06-06T23:14:44.325Z] # skipped 0
[2026-06-06T23:14:44.325Z] # todo 0
[2026-06-06T23:14:44.325Z] # duration_ms 44.855458
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Frontend Build)
[Pipeline] tool
[Pipeline] envVarsForTool
```

Captura 5. Consola Jenkins: docker ps, docker compose build e imagenes Built

Consola Jenkins - Docker habilitado y contenedores contruidos

```
[2026-06-06T23:13:56.188Z] + docker ps
[2026-06-06T23:13:56.189Z] CONTAINER ID   IMAGE                                COMMAND                  CREATED    STATUS    PORTS
[2026-06-06T23:13:56.189Z] 1928b89645ff  skillcert-jenkins:local             "/usr/bin/tini -- /u..." 17 seconds ago  Up 17 seconds  0.0.0.0:8080
[2026-06-06T23:14:46.886Z] + docker compose build
[2026-06-06T23:14:49.735Z] Image skillcert_ci_1-server Built
[2026-06-06T23:14:49.735Z] Image skillcert_ci_1-client Built
[2026-06-06T23:14:50.081Z] SkillCert CI pipeline completed successfully.
Finished: SUCCESS
```

Captura 6. Jenkinsfile publicado en GitHub



Solucion del error de Docker

El error de permisos al Docker local se resolvió dando al agente Jenkins acceso al daemon Docker. En esta evidencia se usó Jenkins en contenedor con el socket Docker montado y usuario root:

```
docker run -u root -v /Users/johnvanegas/.docker/run/docker.sock:/var/run/docker.sock skillcert-jenkins:local
```

En un agente Linux tradicional, la solución equivalente es: `sudo usermod -aG docker jenkins && sudo systemctl restart jenkins.`

Webhook GitHub

La URL requerida para GitHub es `http://SERVIDOR_JENKINS/github-webhook/`. La captura real del webhook no se pudo completar desde este entorno porque GitHub solicito sesion/permisos de administrador al abrir Settings > Webhooks. No se incluye una captura falsa; debe completarse desde la cuenta propietaria del repositorio.

Anexo 1: Jenkinsfile completo

```
pipeline {
    agent any

    tools {
        nodejs 'NodeJS-20'
    }

    options {
        timestamps()
        disableConcurrentBuilds()
        buildDiscarder(logRotator(numToKeepStr: '10'))
    }

    environment {
        NODE_ENV = 'test'
        COMPOSE_PROJECT_NAME = "skillcert_ci_${env.BUILD_NUMBER}"
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/Jonevan369/skillcert-integracion-continua'
            }
        }
    }
}
```

```
stage('Install Dependencies') {

    steps {

        sh 'node --version && npm --version'

        sh 'npm ci'

    }

}

stage('Backend Tests') {

    steps {

        sh 'npm test -w server'

    }

}

stage('Frontend Build') {

    steps {

        sh 'npm run build'

    }

}

stage('Validate Docker Compose') {

    steps {

        sh 'docker ps'

        sh 'docker compose config'

    }

}

stage('Build containers') {

    steps {

        sh 'docker compose build'

    }

}

}
```

```

post {
    always {
        sh 'docker compose down --volumes --remove-orphans || true'
    }
    success {
        echo 'SkillCert CI pipeline completed successfully.'
    }
    failure {
        echo 'SkillCert CI pipeline failed. Review the failing stage logs in Jenkins.'
    }
}
}

```

Anexo 2: Extracto de consola Jenkins

```

[Pipeline] envVarsForTool

[Pipeline] withEnv

[Pipeline] {
[Pipeline] sh

[2026-06-06T23:14:44.324Z] + npm test -w server

[2026-06-06T23:14:44.324Z]

[2026-06-06T23:14:44.324Z] > @competence-platform/server@0.1.0 test

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[2026-06-06T23:14:44.325Z] ---

[2026-06-06T23:14:44.325Z]   duration_ms: 1.658542

```



```
[2026-06-06T23:14:44.325Z]    ...

[2026-06-06T23:14:44.325Z] # Subtest: calculateKarmaDelta rewards consensus alignment and penalizes
disagreement lightly

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[2026-06-06T23:14:44.325Z]    duration_ms: 0.11425

[2026-06-06T23:14:44.325Z]    ...

[2026-06-06T23:14:44.325Z] 1..3

[2026-06-06T23:14:44.325Z] # tests 3

[2026-06-06T23:14:44.325Z] # suites 0

[2026-06-06T23:14:44.325Z] # pass 3

[2026-06-06T23:14:44.325Z] # fail 0

[2026-06-06T23:14:44.325Z] # cancelled 0

[2026-06-06T23:14:44.325Z] # skipped 0

[2026-06-06T23:14:44.325Z] # todo 0

[2026-06-06T23:14:44.325Z] # duration_ms 44.855458

[Pipeline] }
```

Conclusiones

La integración continua quedo aplicada y demostrada con Jenkins: el job clono el repositorio, instalo dependencias, ejecuto pruebas, compilo el frontend, valido Docker Compose y construyo los contenedores. La ejecución final termino en SUCCESS y quedo soportada con capturas y consola exportada.

Referencias

- Jenkins. (2026). Pipeline. Jenkins User Documentation.
<https://www.jenkins.io/doc/book/pipeline/>
- Docker. (2026). Docker Compose overview. Docker Docs.
<https://docs.docker.com/compose/>
- GitHub. (2026). Webhooks documentation. GitHub Docs.
<https://docs.github.com/en/webhooks>